

Clinical Risk Assessment Report: Patient 95

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 43.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.95x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +5.0%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.37 [Bottom 8%] (+5.0%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Patient Age (Age) **Value: 41.0 [Bottom 1%] (+3.6%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.56 [Avg (69th %ile)] (+2.7%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.83 [Top 11%] (+2.5%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.12 [Top 12%] (+2.3%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.7 [Avg (69th %ile)] (+2.2%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.27 [Bottom <1%] (+2.1%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Carcinoembryonic Antigen (CEA) **Value: 8.95 [Top 11%] (+2.0%)**
Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

